

Education

Yale University

Candidate for BS in Mechanical Engineering (ABET), Varsity Athlete (Crew)
Noble and Greenough School
Highest Distinction

2024 - Present

GPA: 4.0

2018 - 2024

GPA: 11.0/11.0

Skills

Design & Analysis: NX, Teamcenter, SolidWorks, 3DX, OnShape, GD&T, ANSYS Workbench, Mechanical, Fluent, CFX

Programing: Python, Java, VBA, Excel API, SolidWorks API, MATLAB, Arduino, Ardupilot, Px4

Manufacturing: Lathe, Mill, Shop machines, Power tools, Soldering, FDM, SLA, SLM, MJF printing

Work Experience

SPACEX | Starbase, TX

June 2026 - September 2026

Booster Vehicle Design Intern

- Worked across analysis and manufacturing teams to upgrade booster capability and reliability
- Designed upgraded capability sheet metal joiners and stringers for gross acreage, fuel tank door, and grid fins, working with analysis teams to reduce shear, buckling, and thermally induced failure modes.
- Coordinated with propulsion and structures teams to integrate raceway DFM and capability upgrades across all tank stations, including ownership of high level assembly releases.

UNITZERO | New Haven, CT

December 2025 - June 2026

Robotics Consultant

- Designed Universal Manipulator Interface (UMI) with Camera, VR tracker, Jetson Nano for VLA training.
- Designed 3D-printed 6-DOF hall-effect based leader arm for robot arm control and data collection.

DRAPER | Boston, MA

June 2025 - August 2025

Mechanism and Electromechanical Design Intern

- Worked across teams to support electronics development and testing for defense applications.
- Automated creation of precision cut and form tooling for quick-turn IC development using parametric modeling and Solidworks API, resulting in auto-generated models and GD&T drawings.
- Developed cooling system data infrastructure and control software for Celestial Navigation System (CNS) using TCP and Modbus.
- Designed and analyzed mass model of CNS system to meet prime contractor requirements, performed fatigue, shock response spectrum, modal, and functional shock analysis using ANSYS and SolidWorks.

Project Experience

YALE UNDERGRADUATE AEROSPACE ASSOCIATION | New Haven, CT

Project Lead, Yale Jet Aeronautics

January 2025 - Present

- Attempting to break the sound barrier with an air-breathing vehicle
- Leading the design, manufacturing, and testing of 350N thrust 2-stage jet engine and corresponding airframe, leveraging SLM and DMLS superalloy printing.
- Turbomachinery design and optimization using CFX.

Aerostructures Lead, Project Liquid

September 2024 - June 2026

- Responsible for airframe design, packaging, and regeneratively cooled liquid rocket, subteam leadership.

FIRST ROBOTICS COMPETITION TEAM 125 - NUTRONS | Revere, MA

December 2017 - June 2024

Team Captain, Team Leader of Design and Systems Integration, Subsystem Design Lead

- Led team to finals of world championship, 12 event wins.
- Owned overall robot integration, coordination between hardware, software, and manufacturing subteams.
- Designed launchers, feeders, drivetrains, telescoping arms. Performed handcalcs, FEA, led CAD.

PERSONAL PROJECTS

Hybrid VTOL

August 2023 - December 2024

- Designed and built patent-pending wingeron/tricopter hybrid capable of plane-like forward flight, efficient hover, and decoupled pitch & translation. Successfully evaluated forward and tricopter flight modes.
- Wrote custom flightcontrol software for Teensy microcontroller.

High-Efficiency FPV Racing Wing

October 2023 - January 2024

- Designed, optimized, and built FPV racing wing capable of 102mph flight on only 3s LiPo powertrain.
- Used Xflr5 for airfoil, sweep, and wash optimization, longitudinal stability analysis.

Extracurricular Activities and Awards

- National Merit Scholarship Finalist
- First Robotics Einstein World Championship Finalists

2024

2023